A study of viscous flow regimes in a large-eddy simulation

Alec Stanley

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1 Introduction

Large eddy simulation (LES) is a computational fluid dynamics (CFD) technique used to balance detailed and complex turbulent flows with more manageable computing demands ([1]). This report uses the publicly accessible data provided by Towne 2023 [2] using the CharLES solver to investigate viscous flow regimes over a NACA0012 airfoil at a six degree angle of attack. The simulation was performed at a Reynolds number of 23,000 and a Mach number of 0.3, consisting of a 2D slice of a 3D simulation.

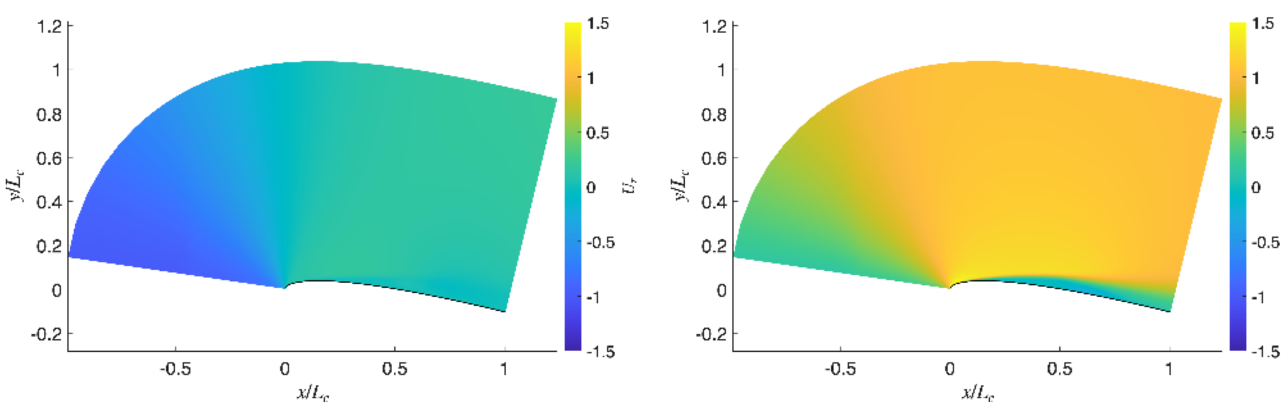
This study aims to identify the relevant flow regimes present, examine boundary layer behaviour, and analyse the turbulent wake behind the airfoil. This is done to demonstrate a thorough understanding of the relevant aerodynamic principles.

2 Results and Discussion

2.1 Flow Regime Identification

The data obtained by Towne 2023[2], referred to in this report as "the database" uses a blended polar coordinate system. To be explicit, the database provides an r-velocity and a t-velocity component. In standard aerodynamics it is typical to use the velocity component in the direction normal to the airfoil surface, and the component in the direction tangent to the airfoil surface (U_n and U_t respectively). This is a more general form of a typical polar system where there is a radial velocity, and a 'θ' velocity running clockwise perpendicular to the radial velocity. The database provides the velocity normal to the airfoil surface as the 'radial' velocity U_r (equivalent to U_n in this case, to avoid confusion) and a standard tangential velocity, while working in a surface-relative coordinate system where the radius is the distance from the closest point on the airfoil, and θ is the angle between that normal vector and the positive x-axis.

In standard laminar flow over a flat plate we typically study the boundary layer formation in the positive x-direction, running tangent to the surface. By analogue, the component of the velocity most relevant to the boundary layer flow over the airfoil is the tangential velocity U_t. Figure 1 depicts the normal and tangential velocity at all points above the airfoil.



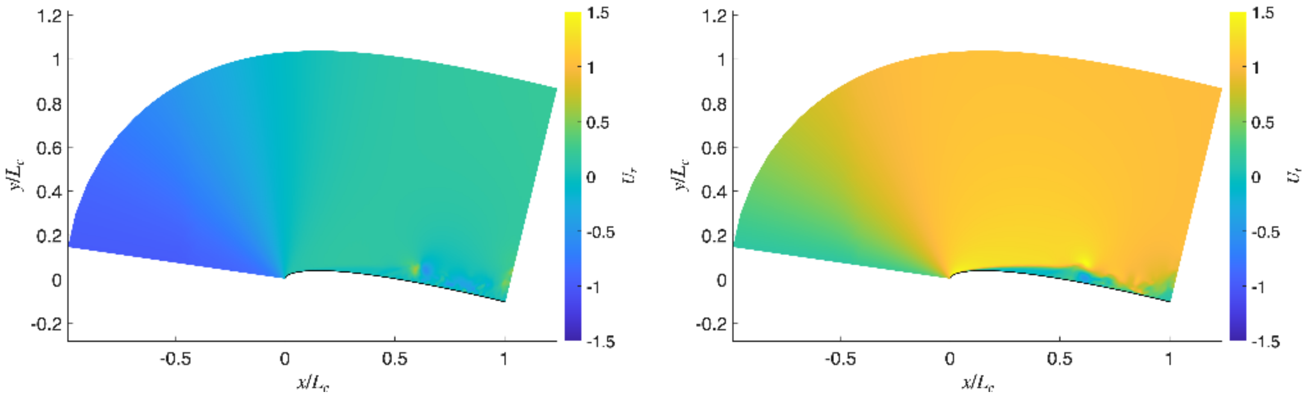
(a) Tangential velocity field

(b) Normal velocity field

Figure 1: The time-averaged surface relative velocity components of the flow over the airfoil.

There are a few interesting artefacts seen in Figure 1. Firstly, there is a very distinct boundary layer forming. This is characterised by a no-slip boundary condition at the airfoil surface (zero velocity at all points on the surface) and a gradual increase in speed until it closely matches the free-stream velocity. Another interesting artefact is that we start to see signs of a possible flow detachment near the trailing edge, as we can see the velocity field starts to increase in magnitude slightly. We will explore this further. Lastly, we also see a stark increase in the air velocity around the stagnation point (leading edge) as the air tightly hugs the airfoil.

The flow detachment is subtle in Figure 1, but is shown much more clearly in Figure 2. This data was taken at an arbitrary point during the simulation (chosen to be timestep 500 of the 1600 time steps), where the assumption is made that the flow was determined to be fully formed before snapshots of the data were taken[2].

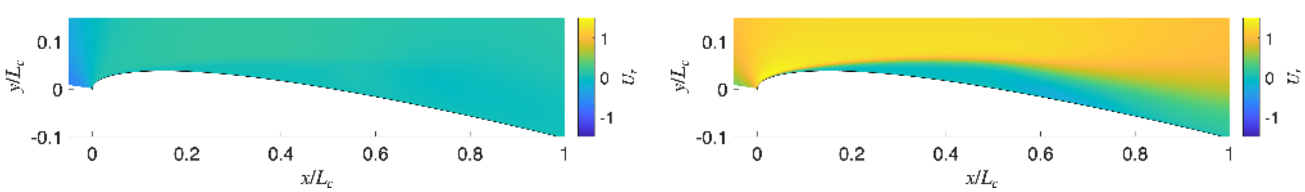


(a) Instantaneous normal velocity field

(b) Instantaneous tangential velocity field

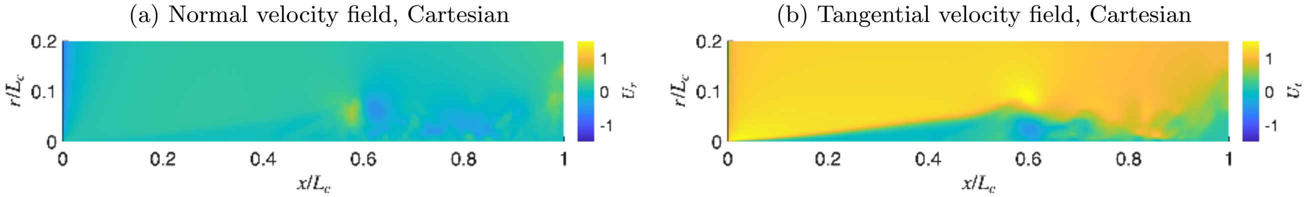
Figure 2: The instantaneous surface relative velocity components of the flow over the airfoil at the 500th simulation time step.

It is highly evident that the flow has detached due to viscous effects, and this is a strong indicator that the flow operates in the turbulent regime. This effect is highlighted in Figure 3 revealing a much clear zoomed in perspective of the boundary layer, complemented by the instantaneous boundary layer data mapped onto the surface of the airfoil.



(a) Normal velocity field, Cartesian

(b) Tangential velocity field, Cartesian

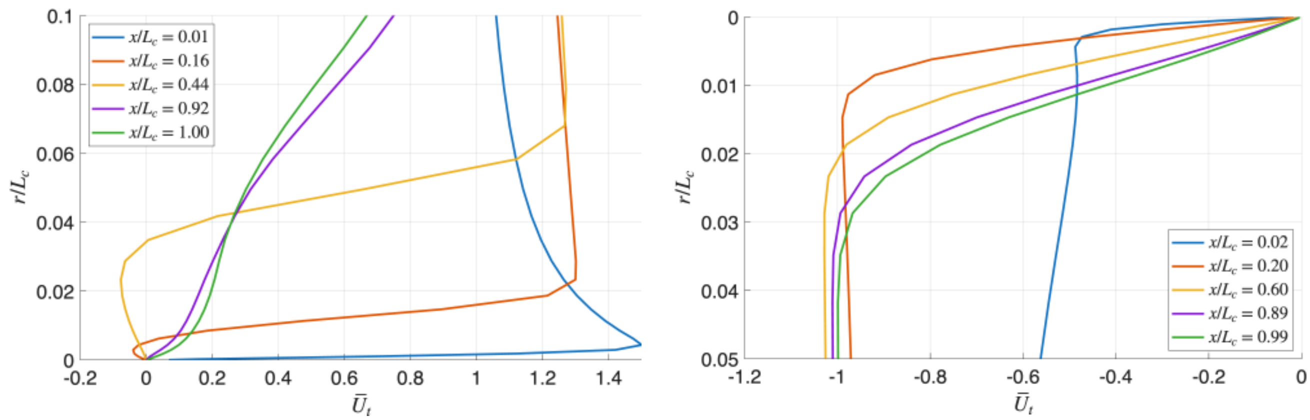


(c) Instantaneous normal velocity field, mapped to airfoil

(d) Instantaneous tangential velocity field, mapped to airfoil

Figure 3: The time-averaged (top) surface relative velocity components of the flow over the airfoil zoomed in to the boundary layer, and the respective instantaneous (bottom) velocities mapped to the airfoil surface.

To understand the different flow regimes across the airfoil we can examine the tangential velocity profile across the surface at various spanwise sections. Figure 4 depicts the profile for various stations above and below the airfoil. The upper airfoil velocity profiles clearly reinforce the presense of flow detachment. In all cases, the no-slip boundary condition is observed. In the profiles closer to the stagnation point, the flow is moving fast enough such that there is enough shear force to overcome the viscous forces. What we see is a very clear steady boundary layer in the laminar regime. The velocity increases away from the surface and eventually stabilises to the freestream velocity. Conversely, the stations closer to the trailing edge are characterised by detached transitional flow followed by the turbulent regime. We can see the velocity immediately dips negative away from the surface before slowly being pulled back to the free stream. This is an expected result[3] where an adverse pressure gradient arising from viscous effects is enough to overcome the flow shear stress, causing the flow to physically detach from the surface.



(a) Upper airfoil surface

(b) Lower airfoil surface

Figure 4: The time-averaged streamwise velocity profiles at various stations along the upper (left) and lower (right) airfoil surfaces.

Another useful metric is that which is obtained by taking the root-mean-square of the mean-subtracted tangential velocity.

$$U'_{t,\mathrm{RMS}} = \mathrm{RMS}(U_t - \bar{U}_t) \tag{1}$$

The field for this velocity metric is given in Figure 5.

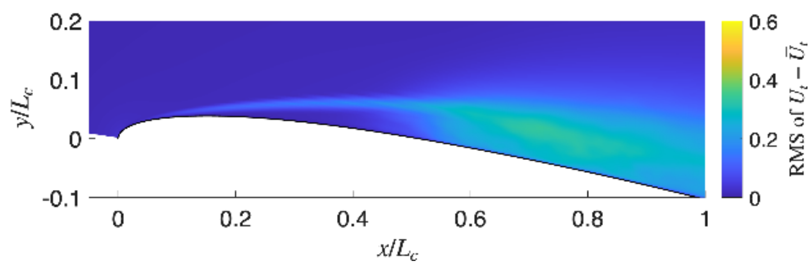
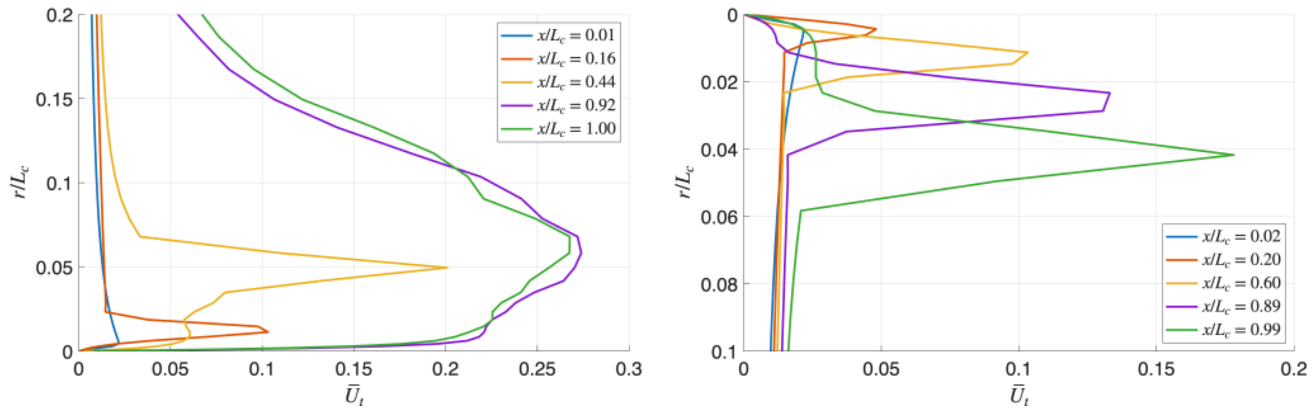


Figure 5: The root-mean-squared of the mean-subtracted velocity field.

This is a highly useful metric as it directly shows different flow regimes. In the laminar regime, the average velocity will be very close to the instantaneous velocity, and $U'_{t,\mathrm{RMS}}$ would be close to zero. Once the flow starts to transition and eventually become turbulent, the instantaneous velocities start to diverge from the mean and we get a noticable $U'_{t,\mathrm{RMS}}$. The RMS field is essentially a map of aerodynamic stress, where high values indicate strong turbulent shear stress (Reynolds stresses). Figure 5 very clearly shows a transition region of flow detachment approximately between 0.5 to 0.6 (of chord length), followed by the turbulent regime from 0.6 onwards on the upper surface. On the lower airfoil surface, the flow appears to be laminar across the entire surface.

To further show this phenomenon we can plot the streamwise velocity profiles at various locations along the upper and lower airfoil surface for the RMS field. This is shown in Figure 6.



(a) Upper airfoil surface

(b) Lower airfoil surface

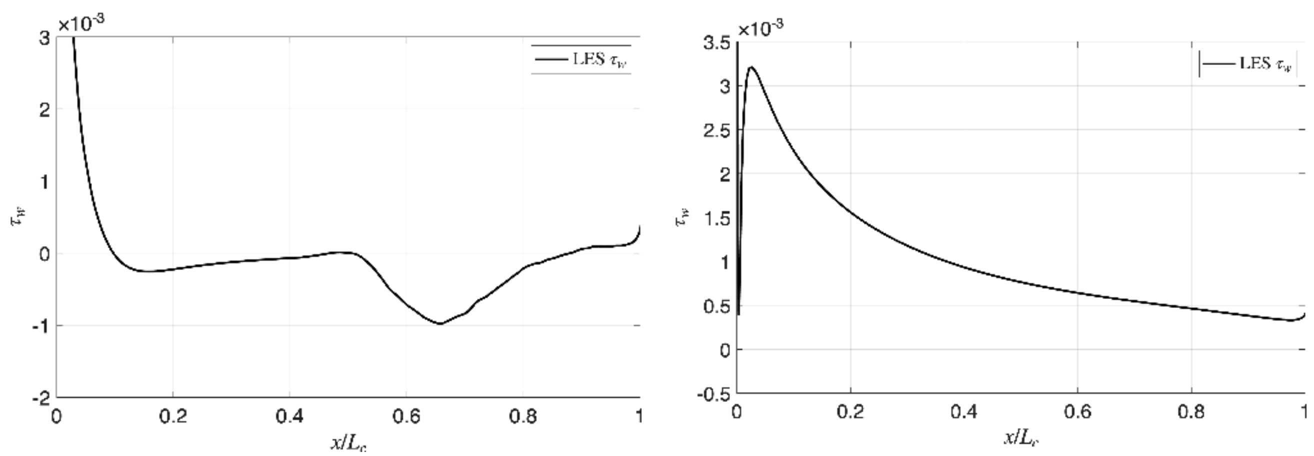
Figure 6: The time-averaged streamwise RMS velocity profiles at various stations along the upper (left) and lower (right) airfoil surfaces.

The earlier streamwise stations exhibit a very low RMS velocity, except for the spike at the edge of the boundary layer where there is a spike in shear stress as the boundary layer meets the free stream. As we surpass the transition we can very distinctly see a much larger RMS velocity, signifying the presence of turbulence. Similar to our previous observations, the flow underneath the airfoil appears to be entirely laminar since the RMS velocity stays very close to zero within the boundary layer edge. The spikes observed at the boundary layer edge indicate the presence of a laminar separation bubble, which will be explored further in this report.

To study the point of flow detachment we can look at the wall shear stress exerted by the flow. A negative shear stress indicates detached flow. The wall shear stress is determined by

$$\tau_w = \rho\nu \left.\frac{du}{dr}\right|_{r=0} . \tag{2}$$

See Figure 7.



(a) Upper surface shear stress

(b) Lower surface shear stress

Figure 7: The shear stress at the airfoil's upper (left) and lower (right) surfaces.

Interestingly, Figure 7 tells us that the flow before the transition is not actually entirely laminar. It is known as the laminar separation bubble (LSB). During this regime, the flow does not have enough momentum to overcome the adverse pressure gradient. It is slightly detached, and then reattaches at $x/L_c \approx 0.45$ before fully detaching at $x/L_c \approx 0.5$. The presence of the LSB explains why we saw a thin turbulent mixing layer right at the top edge of our boundary layer in the RMS velocity field.

We can also observe from this plot that the flow below the airfoil appears to be entirely laminar. The lower surface does not experience an adverse pressure gradient due to the airfoil's angle of attack, where the flow is directly hitting the surface. As a result, we do not see a laminar separation bubble, a flow detachment, or a transition to turbulence as we see on the upper airfoil surface.

2.2 Boundary Layer Effects

In the case of flow over an airfoil at a positive angle of attack, we would expect the upper flow to experience a very strong adverse pressure gradient, fundamentally changing the shape of the boundary layer in contrast to flow over a flat plate (where there is no pressure gradient). We would also expect the shape of the boundary layer to change at the point of separation. The formation of the wake would induce flow recirculation, physically expanding the boundary layer. We can visually see this effect in Figure 8.

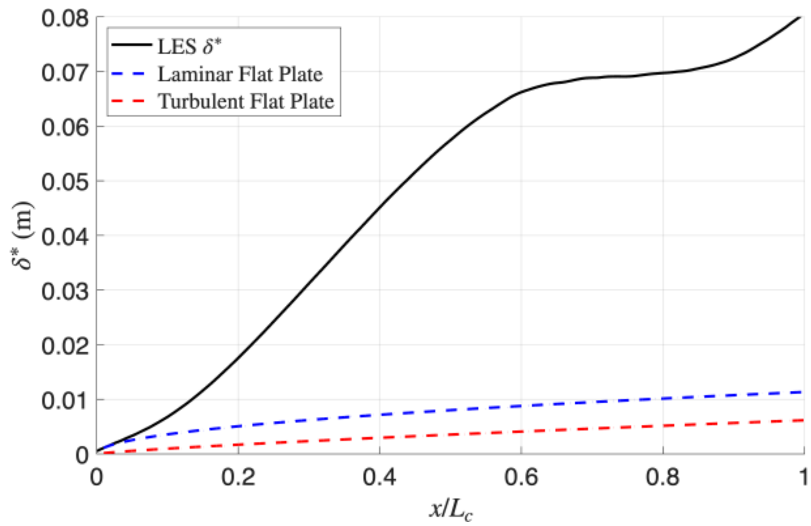


Figure 8: The displacement thickness of the flow over the upper airfoil, plotted against theoretical laminar and turbulent boundary layer shapes (see Appendix).

As expected, the airfoil boundary layer is much larger than the flat plate boundary layers. We also see the shape of the boundary layer physically change shape after separation at the point of flow recirculation.

We may also non-dimensionalise the flow according to the following non-dimensionalisations, which are useful to compare the flow to established theories. We can set the velocity and y-coordinate such that

$$U^+ = \frac{U}{u_\tau}, \text{ and } y^+ = \frac{u_\tau y}{\nu}, \tag{3}$$

where $u_\tau = \sqrt{\frac{\tau_w}{\rho}}$, specifically chosen such that the adjusted Reynolds number in this system is 1. Figure 9 shows these non-dimensional velocity profiles at various streamwise stations.

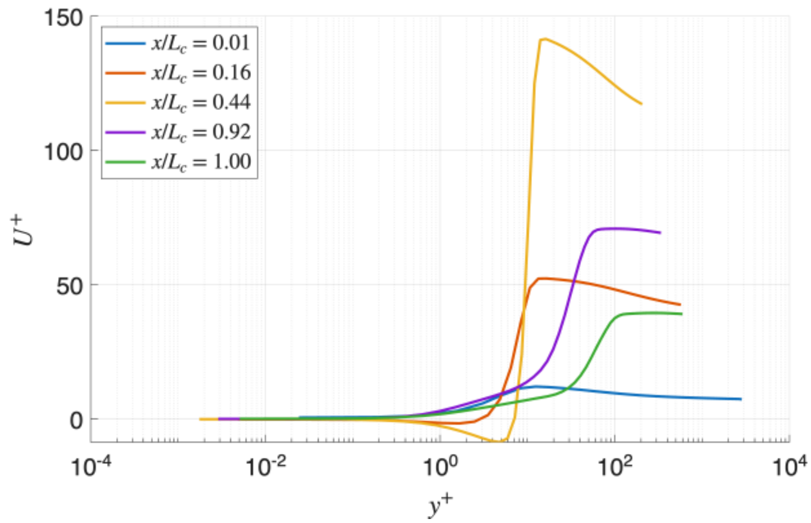


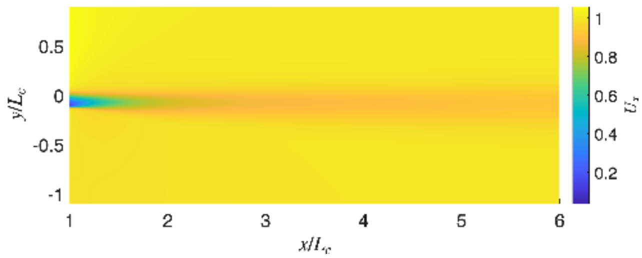
Figure 9: The dimensionless velocity against the dimensionless wall-distance at several streamwise stations along the upper airfoil surface.

In a laminar flow we would expect an approximately linear relation where $U^+ \approx y^+$. In the figure, we can see that earlier stations follow this approximation very closely. As the flow encounters the turbulent transition (seen with the largest yellow line), however, we see a more characteristic logarithmic shape known as the log-law region[3]. After $x/L_c \approx 0.6$, where the flow fully detaches, the profile starts to collapse as the logarithmic assumption breaks down, resulting in a larger deviation in the profile.

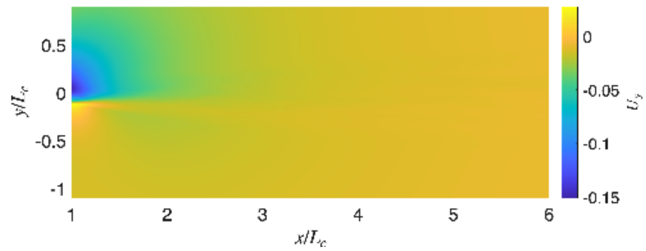
The coefficient of friction for this airfoil was calculated to be 0.0052. Comparing this to the literature, we know the Blasius solution for the skin friction over a laminar flat plate is 0.0088 [3]. The lower friction coefficient is an expected result in this regime due to the presence of a high adverse pressure gradient as well as flow detachment. Both phenomena lower the velocity gradient at the wall, directly reducing skin friction. If we were studying purely laminar flow over an airfoil, however, it would be higher than the flat plate case.

2.3 Wake Profile

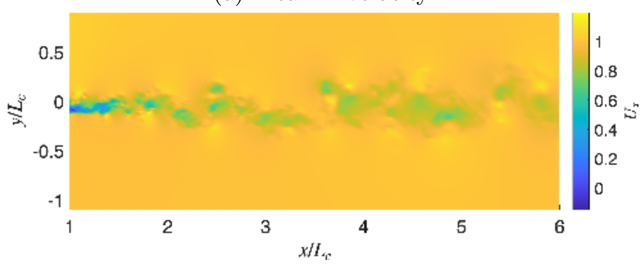
To investigate the airfoil wake we can first look at both the time-averaged and instantaneous velocity fields, similar to Figure 1. The instantaneous snapshots were chosen at an arbitrary timestep in the simulation, since the flow is already assumed to be fully formed. This is shown in Figure 10.



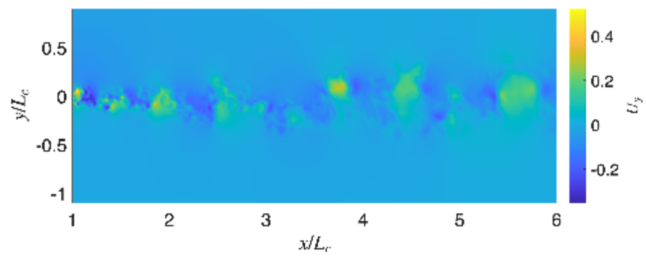
(a) Mean X-velocity



(b) Mean Y-velocity



(c) Instantaneous X-velocity



(d) Instantaneous Y-velocity

Figure 10: The time-averaged (top) and instantaneous (bottom) velocity fields of the airfoil wake in Cartesian coordinates. The arbitrary time step was chosen to be the 500th of 1600 steps.

A very obvious result from the database is the presence of large recirculating eddies in the turbulent wake. We can see the aftermath of the no-slip boundary condition at the left-most data column in all the plots in Figure 10. We then see a gradual return to free-stream conditions as pressure in the wake is slowly recovered. After six chord lengths (the full length of the data given in the database) there is still turbulent mixing occurring in the wake. These wakes are known to persist as far as beyond twelve chord lengths.

When analysing the self-similarity of a boundary layer it is typical to normalise the velocity profile by the boundary layer thickness. When analysing the self-similarity of a turbulent wake, we are looking for some analogous metric that describes the size, or width, of the wake as it moves. The typical measurement used here is the half-width. To find it, we first look at the wake's velocity deficit, the difference between the velocity in the wake and the free-stream velocity. What we should find is that the largest deficit (the slowest flow) occurs where we should define the wake centre. Once we establish the wake centre, we can look at each side of the centreline and obtain a point where the velocity deficit is exactly half the free stream velocity. A helpful comparison can be made to how the boundary layer thickness δ_{99} is defined to be the point where the flow velocity reaches 99% of the free-stream's, here we are looking for the 50% mark (the half maximum). The wake half-width b is the average of the upper and lower half maximum distances. We can construct a plot of the velocity deficit against the y-position normalised by the half width, as seen in Figure 11.

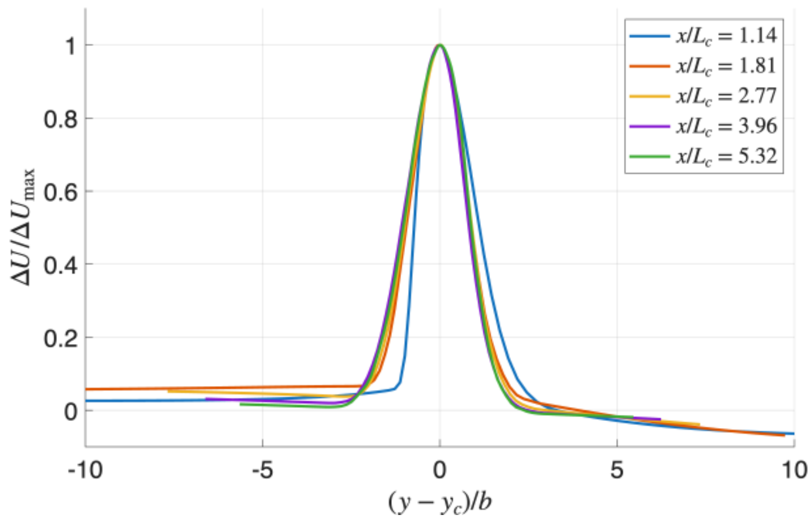


Figure 11: The wake velocity deficit profiles as a function of position normalised by the wake half-width for various streamwise stations across the wake.

A reasonable conclusion to draw from this plot is that the wake is self-similar, meaning it holds its shape for the course of its existence. Another expected result we see here is the wake shape immediately following the trailing edge in the near-wake region. In this region, the wake is still actively evolving and mixing as the airfoil boundary layers meet at the trailing edge. This is why the blue line is slightly offset to the rest of the stations.

To take this further, we can observe the wake's centreline velocity and compare it to what we might expect to see using laminar theory. To find the laminar theory solution for this case we will use the equation

$$U_{c,\mathrm{lam}} = U_\infty - U_\infty \cdot \frac{C_D}{2} \cdot \sqrt{\frac{\mathrm{Re}_L}{4\pi x}}, \tag{4}$$

which is one of the Blasius solutions for laminar flat plate flow [3]. The drag coefficient is taken to be $\frac{2\theta}{L_c}$ using the known analytical equation for the momentum thickness θ (refer to Appendix). See Figure 12.

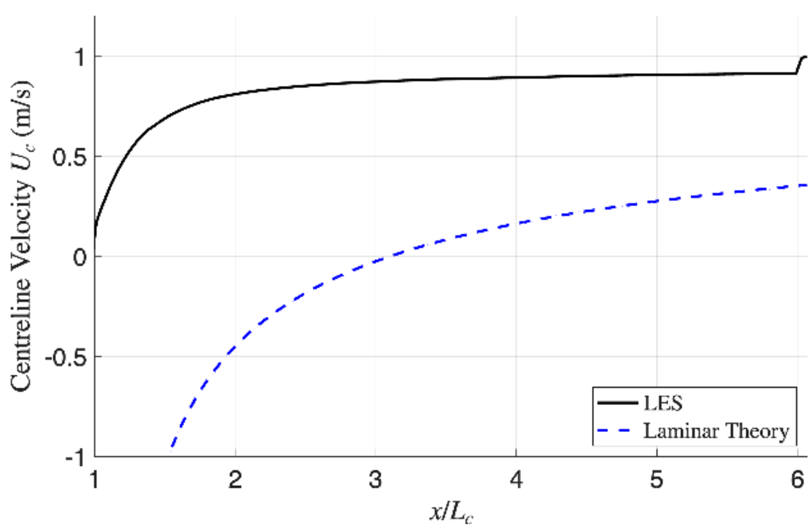


Figure 12: The wake centreline velocity as a function of the chord length, and the expected results from laminar theory

The most striking feature of this graph is that the LES recovers centreline velocity much faster than the laminar case. Such a result is expected, since the extreme mixing that occurs in the turbulent regime would allow pressure and free-stream conditions to be restored much faster. Another artefact is we can clearly see one of the LES boundary conditions, the out-flow must match the free stream velocity. This causes a stark jump right at the end of our data where the velocity must return to the free-stream velocity. We can also discuss the unphysical result that laminar theory predicts a negative velocity in the near-wake region. This is a well-known limitation of forming simplified solutions to Navier-Stokes, where the approximation used for laminar centreline wake velocity is defined under the assumption of far-wake conditions[3]. This causes it to break down in the near-wake and cause wildly unphysical reverse flows approaching velocities of $-\infty$ due to a division by zero. Nevertheless the conclusion can be drawn that the turbulent wake recovers much faster than the laminar case.

Figure 13, obtained from Towne 2023[2] plots the power spectral density (PSD) against the non-dimensional frequency. The PSD measures the distribution of kinetic energy across the flow's eddies. The plot shows that there is higher energy concentrated at lower frequencies, followed by a sharp decline in energy as frequency increases. This is called the turbulent energy cascade. The broad peaks in the lower frequencies are the largest energy-containing eddies formed directly from the unstable vortex shedding that comes from the boundary layer separation over the airfoil.

In-class theory can be applied to this plot as the cascade demonstrates the flow of energy from large to small vortex structures[4] (Week 11 content). However this only takes us down to the resolution of the LES solver used to create such a plot. LES only resolves large-scale structures, meaning the physical behaviour of the higher frequency eddies is dominated by simulation artefacts (i.e. interpolation) rather than actual theory.

3 Conclusion

The viscous flow regimes over the airfoil data from the LES database were successfully analysed. It was demonstrated that the lower airfoil surface remained fully laminar, whereas the upper surface consisted of laminar flow (more specifically the laminar separation bubble), a transition to the turbulent regime, and clear flow detachment. Analysis of the RMS velocity field, the wall shear stress, and the displacement thickness of the boundary layer all support these results. It was found that the turbulent boundary layerr deviated from flat-plate theory primarily due to the presence of an adverse pressure gradient and the formation of high-energy eddies. The logarithmic layer was also shown through non-dimensional analysis of streamwise velocity profiles. It was also determined that the turbulent wake was self-similar and recovered to free-stream conditions faster than the laminar theory.

4 References

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5 Appendix

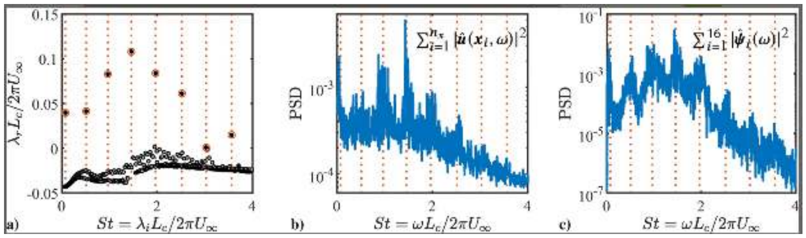

Figure 13: Figure 32c. taken from Towne 2023, representing the power spectral density of flow with respect to the Strouhal number, the non-dimensional frequency.

5.1 Blasius solutions

$$C_f = \frac{1.328}{\sqrt{\mathrm{Re}_L}} \tag{5}$$

$$\delta^*_{\mathrm{lam}} = \frac{1.72x}{\sqrt{\mathrm{Re}_x}} \tag{6}$$

5.2 Other analytical equations (assume incompressible)

$$\delta^* = \int_0^\infty (1 - \frac{u}{u_\infty})dy \tag{7}$$

$$\theta = \int_0^\infty \frac{u}{u_\infty}(1 - \frac{u}{u_\infty})dy \tag{8}$$

$$\delta^*_{\mathrm{turb}} = \frac{0.046x}{\mathrm{Re}_x^{0.2}} \tag{9}$$